

Oakleigh Wilson

University of the Sunshine Coast
School of Science, Engineering, and Technology

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Research Interests

I am a quantitative ecologist integrating big-data computation into ecology. I am interested in animal behaviour and have a life-long passion for evolutionary psychology. My PhD focuses on developing user-friendly theoretical frameworks and data workflows for utilising machine learning in behaviour recognition using animal-borne accelerometers.

Education and Employment

PhD Candidate	Jul 2023 -	<u>University of the Sunshine Coast</u> . Optimising the use of machine learning in animal accelerometer behaviour recognition. Advisors: Christofer Clemente, David Schoeman.
Graduate Ecologist	Jan 2023 - Jul 2024	<u>Eco Logical Australia</u> . Designing data quality procedures and conducting statistical analyses for government and private sector clients.
Research Assistant	Jan - Nov 2021	<u>University of Queensland</u> . Beta testing procedures and running experiments on participants in psychology Perception research lab.
Science Communication Intern	Jan - Dec 2021	<u>The Daily Post</u> . Fortnightly science communication articles for a regional newspaper, targeted at a general audience.
Bachelor of Advanced Science, Biology (Hnrs)	Feb 2019 - Nov 2022	<u>University of Queensland</u> . Thesis focused on quantifying koala ground-traverse behaviour using GPS and accelerometer behaviour recognition. Advisor: Robbie Wilson.

Awards and Grants

2026	<i>Best Student Presentation Award</i>	Australasian Society for the Study of Animal Behaviour Conference
2026	<i>Journal of Experimental Biology Travel Fellowship (\$5000)</i>	The Company of Biologists
2026	<i>Conference Attendance Grant (\$1200)</i>	Association for the Study of Animal Behaviour
2026	<i>Highly Commended Student Research</i>	Australasian Society for the Study of Animal Behaviour
2025	<i>Runner Up Best Student Presentation Award</i>	Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference
2025	Round 1: <i>3MT 1st & People's Choice</i> Round 2: <i>3MT 3rd & People's Choice</i>	University of the Sunshine Coast.
2025	<i>Best Student Presentation Award</i>	Australasian Society for the Study of Animal Behaviour Conference
2025	<i>Conference Travel Grant (\$400)</i>	Ecological Society of Australia

2025	<i>Conference Travel Grant</i> (\$800)	Society of Experimental Biology
2024	<i>Student Research Grant</i> (\$1500)	Ecological Society of Australia
2024	<i>Student Research Grant</i> (\$2500)	Australasian Society for the Study of Animal Behaviour
2023	<i>People's Choice Poster Award</i>	Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference
2023	<i>Runner Up Poster Award</i>	University of the Sunshine Coast School of Science, Engineering, and Technology Showcase
2021	<i>Evolution Prize</i> , highest achievement	University of Queensland School of Biology
2021	<i>Summer Research Scholarship</i> (\$3600)	University of Queensland School of Psychology
2020	<i>Summer Research Scholarship</i> (\$3600)	University of Queensland School of Biology

Publications

Conservation gains at minor movement costs: Evaluating a cat-borne predation deterrent. **O Wilson**, N Galea, D Schoeman, J Annett, C Clemente. (2026) *Applied Animal Behaviour Science*.
<https://doi.org/10.1016/j.applanim.2026.107097>

Ignoring What We Don't Know in Accelerometer-based Behaviour Classification: The Open Set Recognition Problem. **O Wilson**, D Schoeman, B Ferdinandy, C Clemente. (2026) *Journal of Experimental Biology*.
<https://doi.org/10.1242/jeb.250722>

Practical guidelines for validation of supervised machine learning models in accelerometer-based animal behaviour classification. **O Wilson**, D Schoeman, A Bradley, C Clemente. (2025) *Journal of Animal Ecology*.
<http://doi.org/10.1111/1365-2656.70054>

Between the Trees: Quantifying Koala Ground Movement for Conservation. G Sparkes, **O Wilson**, W Ellis, S FitzGibbon, B Barth, C Clemente, M Crowther, R Wilson. (2025) *Animals*.
<https://doi.org/10.3390/ani15243537>

In review

Refining Accelerometer-based Animal Behaviour Classifications with Sequence-Informed Post-Processing. **O Wilson**, H Yu, D Schoeman, C Clemente.

Teaching

2025	Tutorial Lead	SEC602 Ethics in Cybersecurity (University of the Sunshine Coast)
2025 -	Teaching Assistant	ANM100 Animal Ecology Field Course (University of the Sunshine Coast)
2024 - 2025	Teaching Assistant	ANM201 Animal Ecophysiology (University of the Sunshine Coast)
2022	Teaching Assistant	BIOL3205 Animal Ecophysiology (University of Queensland)

Invited Seminars

2025	<i>Fitbits for Wildlife: A Biologging Workflow</i> . Invited Workshop. Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, University of the Sunshine Coast.
2024	<i>Artificial intelligence, human error: Thinking critically about machine learning</i> . Invited Talk. The School of Biomedical Sciences, University of Queensland.

Presentations

2026	Presentation	<i>Behaviour-integrated dead-reckoning: Tracking how wild animals move.</i> Australiasian Society for the Study of Animal Behaviour Conference, Adelaide (Australia).
2025	Presentation	<i>Effect of Body Size on Accelerometer-derived Animal Energy-use Estimates.</i> Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, Sunshine Coast (Australia).
2025	Presentation	<i>From Prediction to Interpretation: Post-Processing to Refine Behavioural Classifications from Animal Accelerometry.</i> Ecological Society of Australia Conference, Adelaide (Australia).
2025	Poster	<i>Practical guidelines for validation of supervised machine learning models in accelerometer-based animal behaviour classification.</i> Society of Experimental Biology Conference, Antwerp (Belgium).
2025	Presentation	<i>Ignoring what we don't know in animal behaviour classification.</i> Australiasian Society for the Study of Animal Behaviour Conference, Armidale (Australia).
2025	Poster	<i>A Biologist's Guide To Machine Learning - Accelerometer Animal Activity Recognition Explained.</i> Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, Gatton (Australia).

Volunteering

2025 -	Committee member & then Co-Chair Media Committee. Society for Open, Reliable, and Transparent Ecology and Evolutionary biology (SORTEE). Here I independently launched and produced an interview science communication series: ' ECR Asks ' where I interviewed researchers around the world on topics related to challenges in academia and open science. Produced monthly blog articles and social media content.
2025 -	Website Assistant & then Website Manager. Australiasian Society for the Study of Animal Behaviour (ASSAB).
2021-2023	Secretary & then President. University of Queensland Social Running Club (UQ SRC).